

Overview

Probability models uncertainty; statistics summarizes and interprets data behavior. AI systems depend on both for risk-aware decision making, evaluation, and monitoring.

Titanic dataset is compact, interpretable, and useful for conditional probability practice.

Core Probability Concepts

- **Sample space** Ω : all possible outcomes.
- **Event** A : subset of outcomes.
- **Probability** $P(A)$: value in $[0, 1]$.

Marginal, Joint, Conditional Probability

- Marginal: $P(A)$
- Joint: $P(A \cap B)$
- Conditional:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

Independence vs Dependence

Events are independent if:

$$P(A \cap B) = P(A)P(B)$$

Otherwise, observing one event changes belief about the other.

Core Statistics Concepts

- Mean:

$$\mu = \frac{1}{n} \sum_i x_i$$

- Variance:

$$\sigma^2 = \frac{1}{n} \sum_i (x_i - \mu)^2$$

- Standard deviation: $\sigma = \sqrt{\sigma^2}$
- Median and quantiles for robust center/spread.

Common distributions (high-level): Bernoulli, Binomial, Normal.

Probability & Statistics on Titanic

Questions: - Baseline survival probability $P(\text{survived})$. - Conditional survival: - $P(\text{survived} | \text{sex=female})$ - $P(\text{survived} | \text{pclass} = 1)$ - Distribution shifts in **age** and **fare** between groups.

Common Pitfalls

- Confusing correlation with causation.
- Ignoring missing values and denominator size.
- Reporting global averages without subgroup context.

Business Use Cases

- Risk scoring and triage.
- Resource allocation under uncertainty.
- Prioritization queues based on conditional probabilities.

Business Case Studies & Exceptions

Case 1: Risk-Based Review Queue

Manual review team has limited capacity. Using unconditional event rate under-prioritized high-risk subgroups.

Mitigation: - Use conditional probabilities by cohort. - Calibrate and monitor subgroup rates.

Case 2: Tail-Risk Blindness

Planning based on average costs ignored upper-tail behavior and caused budget overruns.

Mitigation: - Track quantiles and tail metrics. - Segment analysis by population type.

Interview Questions & Answers

1. **Q: Define conditional probability.**
A: Probability of event A given B occurred, equal to $P(A \cap B)/P(B)$ when $P(B) > 0$.
2. **Q: Independence meaning?**
A: Occurrence of one event does not change probability of the other.
3. **Q: Mean vs median with outliers?**
A: Median is more robust; mean is more sensitive to extreme values.
4. **Q: Why variance important in ML?**
A: It quantifies spread/uncertainty affecting model confidence and calibration.
5. **Q: Titanic conditional example?**
A: Compare survival probability by passenger class or sex.
6. **Q: Why avoid causal claims from observational data?**
A: Confounding variables can create spurious associations.
7. **Q: When use quantiles?**
A: For tail-risk thresholds and robust business cutoffs.
8. **Q: How handle missing data in probability estimates?**
A: Inspect missingness mechanism, impute/drop thoughtfully, and report impact.
9. **Q: Value of probabilistic outputs in business systems?**
A: Enables ranking and risk-based decisions instead of binary-only rules.
10. **Q: Effect of class imbalance?**
A: Base-rate skew can distort naive interpretation of metrics and probabilities.